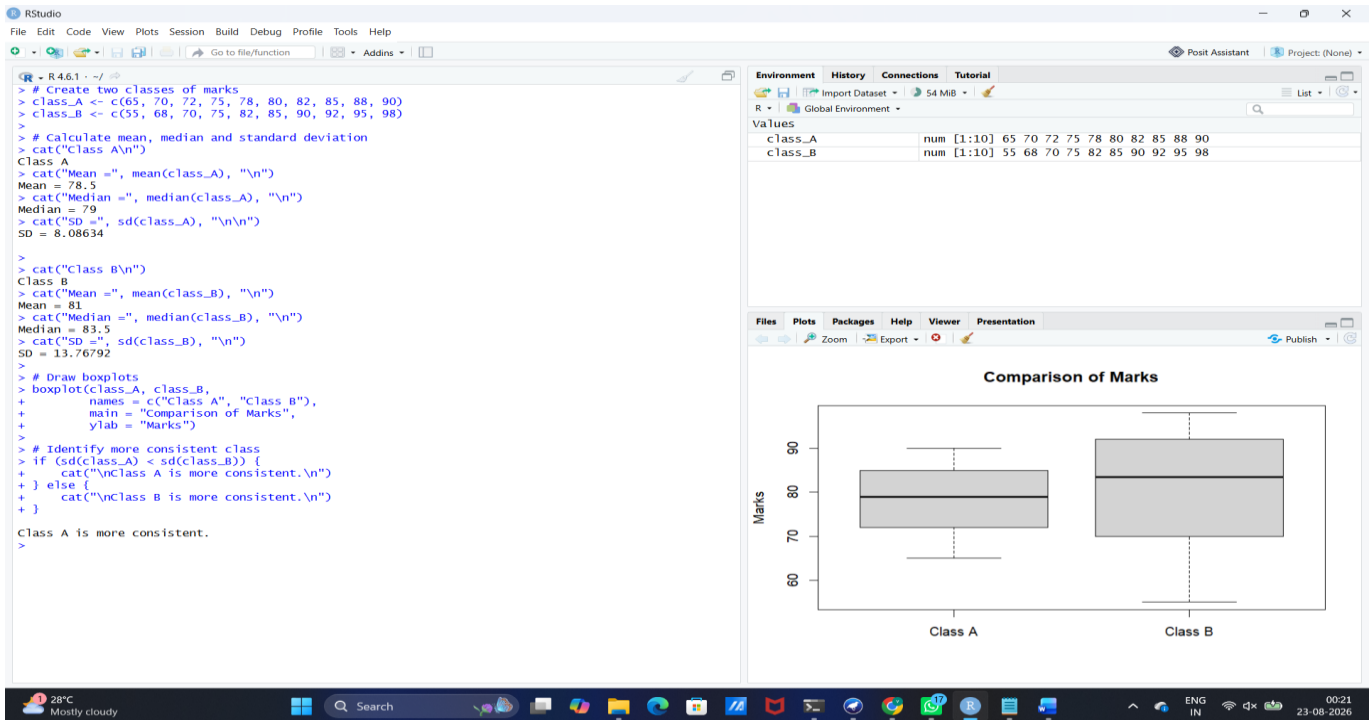
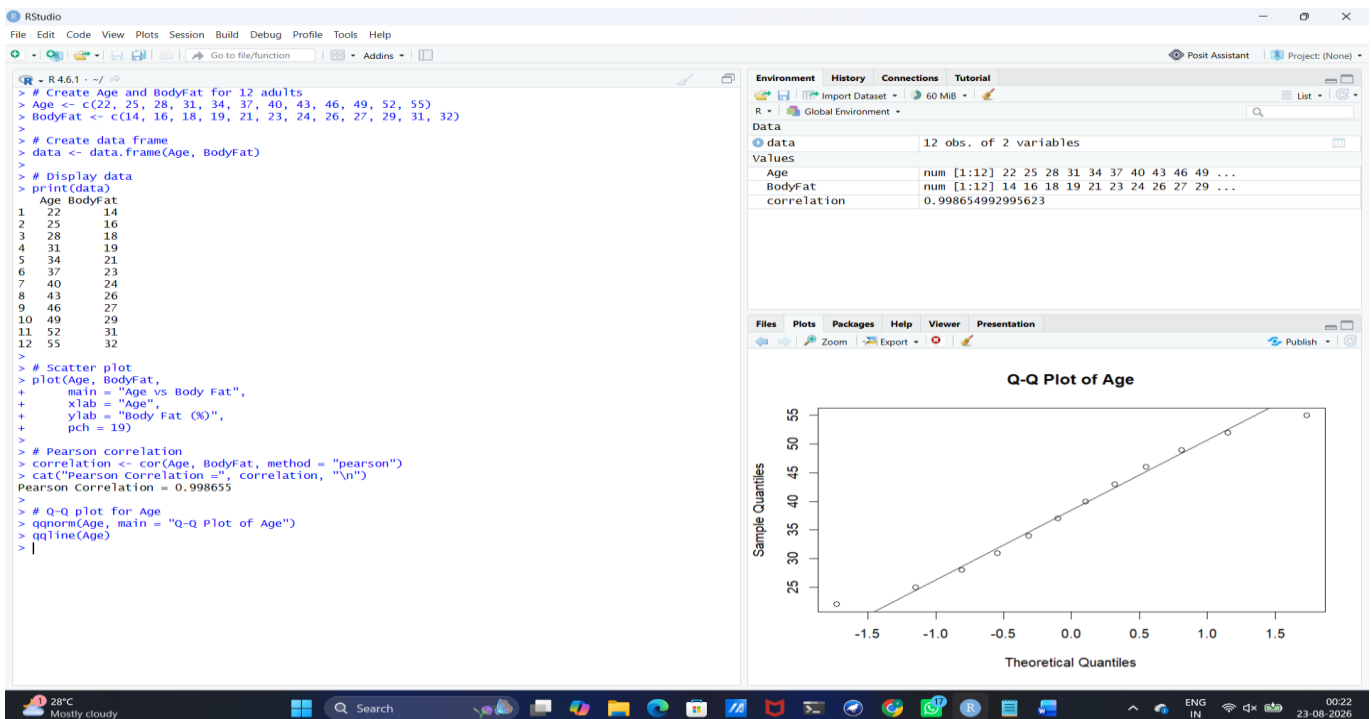


SET 20

1. R: Create two classes of marks; compare means, medians and SDs, draw boxplots and identify the more consistent class.



2. R: Create Age and BodyFat for 12 adults; draw scatter plot, calculate Pearson correlation and create a Q-Q plot for Age.



3. WEKA: Create customer-purchase ARFF data (Age, Income, OnlineVisits, Purchased); compare J48 and SMO using 10-fold CV, accuracy and F1.

Dataset: Customer Purchase Comparison

```
@relation customer_purchase_compare
@attribute Age numeric
@attribute Income numeric
@attribute OnlineVisits numeric
@attribute Purchased {Yes,No}
@data
21,24000,3,No
26,32000,5,No
30,45000,8,Yes
35,52000,10,Yes
41,65000,13,Yes
48,78000,15,Yes
24,29000,4,No
38,58000,11,Yes
55,85000,16,Yes
32,47000,7,No
```

The screenshot shows the Weka Explorer application window. The 'Classify' tab is selected. The classifier chosen is 'J48 -C 0.25 -M 2'. The test options are set to 'Cross-validation' with 8 folds. The result list shows '00:32:27 - trees.J48' as the selected result. The classifier output is displayed on the right, showing stratified cross-validation summary and detailed accuracy by class.

Classifier output

Number of Leaves : 2
Size of the tree : 3
Time taken to build model: 0 seconds

=== Stratified cross-validation ===
=== Summary ===

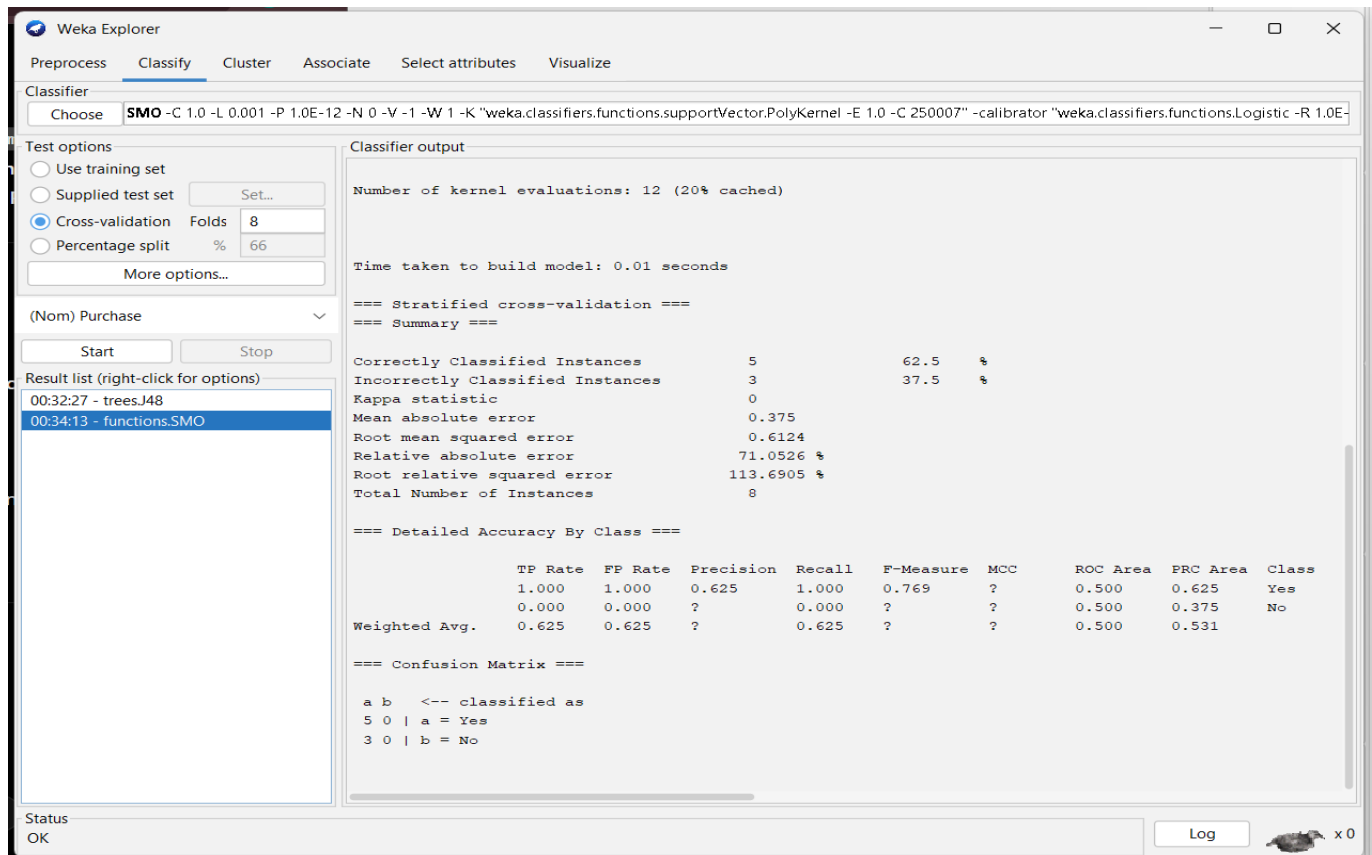
Correctly Classified Instances	7	87.5 %
Incorrectly Classified Instances	1	12.5 %
Kappa statistic	0.7143	
Mean absolute error	0.125	
Root mean squared error	0.3536	
Relative absolute error	23.6842 %	
Root relative squared error	65.6392 %	
Total Number of Instances	8	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	1.000	0.333	0.833	1.000	0.909	0.745	0.833	0.833	Yes
	0.667	0.000	1.000	0.667	0.800	0.745	0.833	0.792	No
Weighted Avg.	0.875	0.208	0.896	0.875	0.868	0.745	0.833	0.818	

=== Confusion Matrix ===

```
a b <-- classified as
5 0 | a = Yes
1 2 | b = No
```



4. WEKA: Create at least 10 transactions; run Apriori and FP-Growth with identical support/confidence, then identify common and unique rules.

Dataset: Transaction Association Comparison

```

@relation market_transactions
@attribute Bread {T,F}
@attribute Milk {T,F}
@attribute Eggs {T,F}
@attribute Butter {T,F}
@attribute Juice {T,F}
@data
T,T,F,F,F
T,T,T,F,F
T,F,T,T,F
T,T,T,T,F
F,T,T,F,T
T,T,F,F,T
F,T,T,T,T
T,F,F,T,T
T,T,T,F,T
F,T,T,T,F

```

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Associator

Choose **FPGrowth** -P 2 -I -1 -N 10 -T 0 -C 0.6 -D 0.05 -U 1.0 -M 0.2

Start Stop

Result list (right-click for ...)

- 00:42:31 - Apriori
- 00:43:48 - FPGrowth

Associator output

```
Butter
Juice

=== Associator model (full training set) ===

Apriori
=====

Minimum support: 0.35 (3 instances)
Minimum metric <confidence>: 0.9
Number of cycles performed: 13

Generated sets of large itemsets:

Size of set of large itemsets L(1): 9

Size of set of large itemsets L(2): 20

Size of set of large itemsets L(3): 11

Best rules found:

1. Butter=F 5 ==> Milk=T 5 <conf:(1)> lift:(1.25) lev:(0.1) [0] conv:(1)
2. Bread=T Butter=F 4 ==> Milk=T 4 <conf:(1)> lift:(1.25) lev:(0.08) [0] conv:(0.8)
3. Eggs=F 3 ==> Bread=T 3 <conf:(1)> lift:(1.43) lev:(0.09) [0] conv:(0.9)
4. Bread=F 3 ==> Milk=T 3 <conf:(1)> lift:(1.25) lev:(0.06) [0] conv:(0.6)
5. Bread=F 3 ==> Eggs=T 3 <conf:(1)> lift:(1.43) lev:(0.09) [0] conv:(0.9)
6. Bread=F Eggs=T 3 ==> Milk=T 3 <conf:(1)> lift:(1.25) lev:(0.06) [0] conv:(0.6)
7. Bread=F Milk=T 3 ==> Eggs=T 3 <conf:(1)> lift:(1.43) lev:(0.09) [0] conv:(0.9)
8. Bread=F 3 ==> Milk=T Eggs=T 3 <conf:(1)> lift:(1.67) lev:(0.12) [1] conv:(1.2)
9. Milk=T Butter=T 3 ==> Eggs=T 3 <conf:(1)> lift:(1.43) lev:(0.09) [0] conv:(0.9)
10. Eggs=T Butter=F 3 ==> Milk=T 3 <conf:(1)> lift:(1.25) lev:(0.06) [0] conv:(0.6)
```

Status OK

Log x0

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Associator

Choose **FPGrowth** -P 2 -I -1 -N 10 -T 0 -C 0.6 -D 0.05 -U 1.0 -M 0.2

Start Stop

Result list (right-click for ...)

- 00:42:31 - Apriori
- 00:43:48 - FPGrowth

Associator output

```
=== Run information ===

Scheme:      weka.associations.FPGrowth -P 2 -I -1 -N 10 -T 0 -C 0.6 -D 0.05 -U 1.0 -M 0.2
Relation:    market_transactions
Instances:   10
Attributes:  5
    Bread
    Milk
    Eggs
    Butter
    Juice

=== Associator model (full training set) ===

FPGrowth found 1 rules (displaying top 1)

1. [Eggs=F]: 3 ==> [Butter=F]: 2 <conf:(0.67)> lift:(1.33) lev:(0.05) conv:(0.75)
```

Status OK

Log x0